

A 0.4–0.9 V Supply Voltage-Flexible Third-Order Passive $\Delta\Sigma$ Modulator With Switched-Capacitor Loop Filter Achieving 71.9 dB Peak SNDR at 4 MHz Bandwidth

Jeroen Ponte^{ID}, Graduate Student Member, IEEE, Harijot Singh Bindra^{ID}, Member, IEEE, and Bram Nauta^{ID}, Fellow, IEEE

Abstract—This paper describes the design and implementation of a 4 MHz bandwidth 3rd-order Discrete-Time (DT) Passive Delta-Sigma Modulator (P $\Delta\Sigma$ M), which avoids the use of supply voltage-sensitive linear amplifiers. The Switched-Capacitor (SC) operation and amplifier-less nature of its loop filter render the proposed modulator highly supply voltage- and clock frequency-flexible. To explain our design choices, we investigate two key design challenges unique to P $\Delta\Sigma$ Ms. Firstly, the noise criticality of the quantizer ADC due to the lack of (voltage) gain in the loop filter preceding it. Secondly, the inaccessibility of the readily synthesizable and highly frequency-selective loop filter frequency responses typically employed in $\Delta\Sigma$ Ms, caused by the absence of linear amplifiers. We show that the noise-power trade-off of the quantizer ADC in 1 bit P $\Delta\Sigma$ Ms is the same as that of the first integrator amplifier in Active $\Delta\Sigma$ M. Next, we cover the suitability of several passive filters for use in P $\Delta\Sigma$ M loops. This leads to the selection of the voltage-sampled Charge-Rotating IIR (CR-IIR) filter, for which we show $\Delta\Sigma$ M-oriented design equations, on which we base the design of our 3rd-order DT P $\Delta\Sigma$ M. Measurement results on the resulting prototype P $\Delta\Sigma$ M, implemented in GlobalFoundries 22 nm FD-SOI CMOS, demonstrate that it achieves a 71.9 dB peak SNDR and 4 MHz bandwidth while consuming 4.4 mW from a 0.9 V supply and occupying 0.076 mm² of silicon area. The prototype P $\Delta\Sigma$ M continues to operate at a 2.25 $\times$ reduced supply voltage of 0.4 V, albeit at a reduced bandwidth of 156.25 kHz, where it achieves a peak SNDR of 62 dB while consuming 37.8 μ W.

Index Terms—Analog-to-digital converter (ADC), discrete-time delta-sigma ($\Delta\Sigma$) modulation, passive switched-capacitor filter, opampless, clock frequency-flexible, supply voltage-flexible, CMOS, 22 nm FD-SOI.

I. INTRODUCTION

THE design of linear amplifiers has become increasingly difficult in modern CMOS technologies due to the reducing supply voltage [1]. New amplifier architectures, such

Received 1 September 2024; revised 5 December 2024 and 22 January 2025; accepted 26 January 2025. This work was supported in part by European Research Council (ERC) under European Union’s Horizon 2020 Research and Innovation Program under Grant 834389 and in part by the National Growth Fund through the Dutch 6G Flagship Project “Future Network Services.” This article was recommended by Associate Editor R. Maurino. (Corresponding author: Jeroen Ponte.)

The authors are with the Integrated Circuit Design Group, University of Twente, 7522 NB Enschede, The Netherlands (e-mail: j.ponte@utwente.nl).
Digital Object Identifier 10.1109/TCSI.2025.3536610

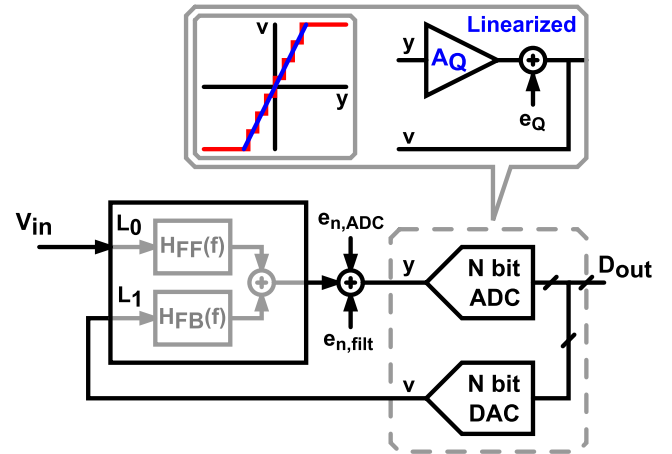


Fig. 1. Generalized model of an N-bit $\Delta\Sigma$ Modulator with $H_{FF}(f)$ and $H_{FB}(f)$ representing the (Continuous- or Discrete-Time) frequency-dependent input and feedback signal transfers of the loop filter. As shown for a multibit example, the quantizer can be modeled by its linearized approximation, replacing its true transfer function (red) with a linear approximation (blue) [4]. Quantizer input-referred and filter output-referred thermal noise are modeled by $e_{n,ADC}$ and $e_{n,flt}$, respectively.

as e.g. the ring amplifier [2], have been proposed to solve this problem. However, an alternative solution is the total elimination of these linear amplifiers from the analog signal chain, as proposed in [3]. This leaves the designer with switches, capacitors, resistors and non-linear gain blocks such as comparators. In our first attempt at this “no gain” approach, we focus our attention on the ADC.

While linear amplifierless Nyquist-rate ADCs (such as e.g. SAR ADCs) are abundant in literature, they require highly frequency-selective active anti-alias filters, which typically employ the linear amplifiers we aim to avoid. Contrarily, the oversampled nature of noise-shaping ADCs relaxes the selectivity requirements on their anti-alias filters, potentially enabling the omission of amplifiers from the latter. Although amplifierless Passive Noise-Shaping SAR ADCs have been presented in literature, the demands on their anti-alias filters are not significantly relaxed due to their low OverSample Ratios (OSRs) [5]. Delta-Sigma Modulators ($\Delta\Sigma$ Ms), on the other hand, commonly utilize high OSRs and are thus a more promising candidate architecture. We therefore wish to

investigate the design and on-chip implementation of a $\Delta\Sigma$ following the no gain approach from [3]. While such Passive $\Delta\Sigma$ s (P $\Delta\Sigma$ s) exist in literature, their low bandwidth of 100kHz [6] limits their application scope. We wish to extend their bandwidth to several MHz, enabling P $\Delta\Sigma$ s for new applications.

A special benefit of P $\Delta\Sigma$ s is that due to the absence of supply-sensitive linear amplifiers, they continue to operate at severely lowered supply voltages, albeit at reduced bandwidth (speed), as we will demonstrate. This could have applications in e.g. IoT sensor nodes, where a MHz bandwidth-enabled supply-flexible ADC could (consecutively) serve a dual role as both radio quantization back-end at full supply voltage, and low-power sensor ADC at reduced supply voltage.

This work discusses the design and implementation of an integrated supply voltage-flexible 3rd-order Discrete-Time (DT) Low-Pass P $\Delta\Sigma$ achieving a 4MHz bandwidth and 71.9dB peak SNDR while consuming 4.43mW from a 0.9V supply. By scaling all core supply voltage down to 0.4V, its power consumption reduces to 37.8 μ W with an associated 156.25kHz bandwidth and 62dB peak SNDR. With these specifications, a single one of our P $\Delta\Sigma$ can be applied for a range of applications in low-power IoT sensors. Additionally, the 62dB SNDR demonstrated at 0.4V operation allow optimizing our P $\Delta\Sigma$ for use in energy harvesting-powered ICs [7].

The text is organized as follows. We begin by discussing two key design challenges that arise in the design of P $\Delta\Sigma$ s, related to the quantizer and loop filter, in Section II. Continuing, Section III goes more in-depth on the quantizer, and Section IV explores loop filter design in P $\Delta\Sigma$ s, culminating in the selection of a filter architecture. Section V derives two design equations to assist in the design of the selected filter, oriented towards P $\Delta\Sigma$ s. Some unique properties of the filter allow our relatively simple equations to cover arbitrary-order (loop) filters. Continuing, Section VI treats the on-chip implementation of our 3rd-order P $\Delta\Sigma$. We apply time-interleaving to the loop filter to increase P $\Delta\Sigma$ bandwidth, accepting the SN(D)R performance penalty resulting from finite filter component mismatch. Standard calibration techniques could reduce the penalty, but this is beyond the scope of this exploratory work. Next, Section VII demonstrates measurement results on our prototype supply-flexible P $\Delta\Sigma$. Its performance at several reduced supply voltages is reported, and far-out-of-band large-signal two-tone measurements are shown. Finally, conclusions are drawn in Section VIII.

II. KEY CHALLENGES IN THE DESIGN OF PASSIVE $\Delta\Sigma$ S

The design of P $\Delta\Sigma$ s presents two key challenges not found in Active $\Delta\Sigma$ s. We begin our discussion by considering Fig. 1, which shows a generalized model of an N bit $\Delta\Sigma$. The loop filter is represented by two (Continuous- or Discrete-Time) frequency-dependent transfer functions $H_{FF}(f)$ and $H_{FB}(f)$. Typically, (different) $H_{FF}(f)$ and $H_{FB}(f)$ are implemented by (differently) coupling the input and DAC feedback signals into a single, shared loop filter. It can be demonstrated that a large (in-band) $H_{FF}(f)$ magnitude, as typically present in active $\Delta\Sigma$ s, suppresses

the input-referred thermal noise contribution of the quantizer ADC $e_{n,ADC}$ [4].

Unfortunately, achieving such large voltage gain in a passive (low-pass) loop filter is challenging. Techniques such as e.g. capacitor stacking can be applied, but this requires additional clock phases and the achievable voltage gain is limited by parasitic loading [8]. Although the quantizer with its (implicit) gain can compensate for the lack of gain in the loop filter, it becomes a key input-referred P $\Delta\Sigma$ noise contributor. As a result, the noise-power trade-off of the quantizer ADC is the first key challenge in the design of P $\Delta\Sigma$. We will first discuss modeling of the (implicit) quantizer gain in Section III-A. Continuing, Section III-B treats the fundamental noise-power trade-off of a 1 bit quantizer ADC in a P $\Delta\Sigma$, and compares it with that of the first linear (integrator) amplifier in Active $\Delta\Sigma$, which plays an analogous role. We show that criticality of quantizer input noise in P $\Delta\Sigma$ is not a direct bottleneck in achieving energy efficiencies similar to their active counterparts.

The second key challenge involves achieving a high modulator resolution in P $\Delta\Sigma$, which requires high quantizer gain, being related to the frequency-selectivity of their loop filter as will be discussed in Section III. In Section IV we treat how the omission of linear amplifiers constrains the loop filter design space, disallowing the implementation of the highly-selective frequency responses typically employed in Active $\Delta\Sigma$.

III. QUANTIZER CONSIDERATIONS FOR PASSIVE $\Delta\Sigma$ S

A. Quantizer Gain Estimation

In order to maximize quantization noise suppression, we must maximize quantizer gain A_Q . To conceptually understand how to do this, we will first model the quantizer using a linear approximation, as illustrated in Fig. 1. Several approaches exist in literature [4], [9]. In this work we will utilize one previously applied in P $\Delta\Sigma$ s, which uses the ratio of the RMS levels of the quantizer output and input signals [9]. We will initially neglect thermal noise, focusing exclusively on quantization noise. This approach allows the designer to sufficiently lower the quantization noise level to achieve a Signal-to-Quantization Noise Ratio (SN_{QR}) greater than their SNR target [4]. This ensures that when thermal noise is introduced, the $\Delta\Sigma$ noise floor is dominated by the latter as is typical for energy-efficient designs [4]. Later in the text we will introduce thermal noise considerations. The spectral energy in the quantizer output signal of a typical correctly-operating $\Delta\Sigma$ with zero input is primarily concentrated around $f_s/2$. Because the quantizer input signal y equals its output signal v filtered by the loop filter feedback response $H_{FB}(f)$, a coarse simplification can be applied [9]

$$A_Q \approx \frac{\text{RMS}\{v\}}{\text{RMS}\{y\}} \approx \frac{1}{|H_{FB}(f_s/2)|} \quad (1)$$

where $|H_{FB}(f_s/2)|$ is the magnitude of the loop filter's feedback response $H_{FB}(f)$ at $f_s/2$. This indicates that the (RMS level of the) quantizer input signal y must be minimized in order to achieve a large A_Q , which can be achieved by maximizing the high-frequency rejection of $H_{FB}(f)$. This can be done either by increasing the order of $H_{FB}(f)$ (i.e. of the loop filter), or by reducing its -3 dB frequency. The latter

choice has an impact on the noise contribution of the quantizer ADC, which will be discussed in Section V.

Besides increasing A_Q , another method of reducing in-band quantization noise is to increase the resolution of the quantizer. However, effective multi-bit quantization of such small y requires closely-spaced levels (i.e. a steep transfer in Fig. 1), which is challenging to implement. Single-bit quantizers, having only two quantization levels, do not suffer from this issue and are thus a more practical choice for P $\Delta\Sigma$ Ms.

B. Quantizer ADC Noise-Power Trade-Off

As discussed, quantizer (input) thermal noise ($e_{n,ADC}$ in Fig. 1) is a key contributor to the input-referred noise of P $\Delta\Sigma$ M, warranting an investigation into its fundamental noise-power consumption trade-off. In the following, we assume that both the loop filter feedback and feedforward responses $H_{FB}(f)$ and $H_{FF}(f)$ have a 0dB in-band gain for the P $\Delta\Sigma$ M, in which case the quantizer ADC's input-referred noise and its contribution to modulator input noise are the same. A situation which is similar for many other conventional ADCs, such as e.g. SAR, pipeline and flash ADCs. The transfer function of this noise contribution to the P $\Delta\Sigma$ M output equals its Signal Transfer Function (STF), and noise outside the P $\Delta\Sigma$ M bandwidth is rejected by the decimation filter. Because the same argument holds for the input-referred noise of the first integrator amplifier in Active $\Delta\Sigma$ Ms [10], it serves as our point of comparison. Finally, we ignore additional power consumption from e.g. clock power in SC filters for the sake of insight.

Due to the impracticality of multi-bit quantization in P $\Delta\Sigma$ M, we only consider the single-bit case, simplifying the quantizer ADC in Fig. 1 to a comparator. For the first integrator amplifier in an Active $\Delta\Sigma$ M, we make two simplifying assumptions. First, we take a simple differential pair to represent its input stage, neglecting noise and power contributions from its (intermediate and) output stage(s). And second, we will not consider the noise contributions of the additional (passive) elements typically surrounding this amplifier, a detailed discussion of which can be found in [11]. Finally, because the supply voltage V_{DD} does not factor into any of the following equations, we will use current consumption as a proxy for power consumption.

We start with the first integrator amplifier, being noise-critical in Active $\Delta\Sigma$ Ms [10]. It was shown in [3] that the supply (tail) current I_{tail} required in a differential pair, consisting of MOS Transistors (MOSTs) operating in weak inversion, for a certain input-referred noise $v_{n,amp,in-band}^2$ integrated across the $\Delta\Sigma$ M bandwidth $BW_{\Delta\Sigma M} = f_{clk}/(2 \cdot OSR)$ is given by

$$I_{tail} = \frac{8(nk_B T)^2}{v_{n,amp,in-band}^2 \cdot q} \cdot BW_{\Delta\Sigma M} \quad (2)$$

where OSR is the OverSample Ratio, n is the MOST subthreshold slope factor, k_B is Boltzmann's constant, T is the absolute temperature in Kelvin and q is the charge of a single electron.

We move on to consider comparators, which are similarly noise critical in 1 bit P $\Delta\Sigma$ Ms. In them, a preamplifier with

sufficient gain is typically applied before the latch stage, designed to suppress the latter's noise [12], [13]. We consider here the dynamic comparator and its analysis from [13], because we utilize its higher-speed variant from [14] in the P $\Delta\Sigma$ M design in this work (shown in Fig. 5), which presents a comparable noise-power trade-off. As the noise contribution of the latch is suppressed by the gain of the preamplifier, the latter dominates overall comparator input-referred noise and current consumption [12], [13]. Because the comparator draws charge packets Q_{DD} from its supply in the reset phase, its average current consumption (over an integer number of clock cycles) can be found as $I_{comp.} = Q_{DD} \cdot f_{clk}$, where f_{clk} is the comparator clock frequency. Using this, the average current consumption of the dynamic comparator from [13] was shown to be related to its total input-referred noise $v_{n,comp,in,total}^2$ (integrated from DC to $f_{clk}/2$) by [3]

$$\overline{I_{comp.}} = Q_{DD} \cdot f_{clk} = \frac{4(nk_B T)^2}{v_{n,comp,in,total}^2 \cdot q} \cdot f_{clk}. \quad (3)$$

Because the digital decimation filter following the P $\Delta\Sigma$ M rejects (the majority of) the noise beyond its bandwidth, only the in-band fraction $v_{n,comp,in-band}^2$ is relevant to our discussion. Assuming that the total noise is white from DC to $f_{clk}/2$, it follows that this fraction equals

$$v_{n,comp,in-band}^2 = \frac{v_{n,comp,in,total}^2}{OSR}. \quad (4)$$

With this result, we can rewrite (3) as

$$\begin{aligned} \overline{I_{comp.}} &= \frac{4(nk_B T)^2}{v_{n,comp,in,total}^2 \cdot q} \cdot f_{clk} \\ &= \frac{8(nk_B T)^2}{v_{n,comp,in-band}^2 \cdot q} \cdot BW_{\Delta\Sigma M} \end{aligned} \quad (5)$$

which is identical to (2), showing that while the quantizer is a key noise contributor in P $\Delta\Sigma$ M, this does not present a fundamental bottleneck in achieving energy-efficiencies similar to their active counterparts.

IV. P $\Delta\Sigma$ M LOOP FILTER TOPOLOGY SELECTION

The relatively low quality factor and large area of on-chip CMOS inductors at baseband frequencies [15] practically limits the choice of practical passive filter elements to Resistors (R), Capacitors (C) and Switched-Capacitors (SC), thus constraining the set of available filter responses. The omission of linear amplifiers (and linear transconductors) means high-Q complex poles cannot be implemented on-chip, prohibiting the use of high-Q resonator-based loop filter responses which are typically employed in Active $\Delta\Sigma$ M [4]. Further restrictions, on which we elaborate in Section V, additionally render impractical the implementation of pure integrator-based loop filter responses. As indicated by (1), our aim in selecting a filter topology using the available passive circuit elements should be to maximize its high-frequency rejection.

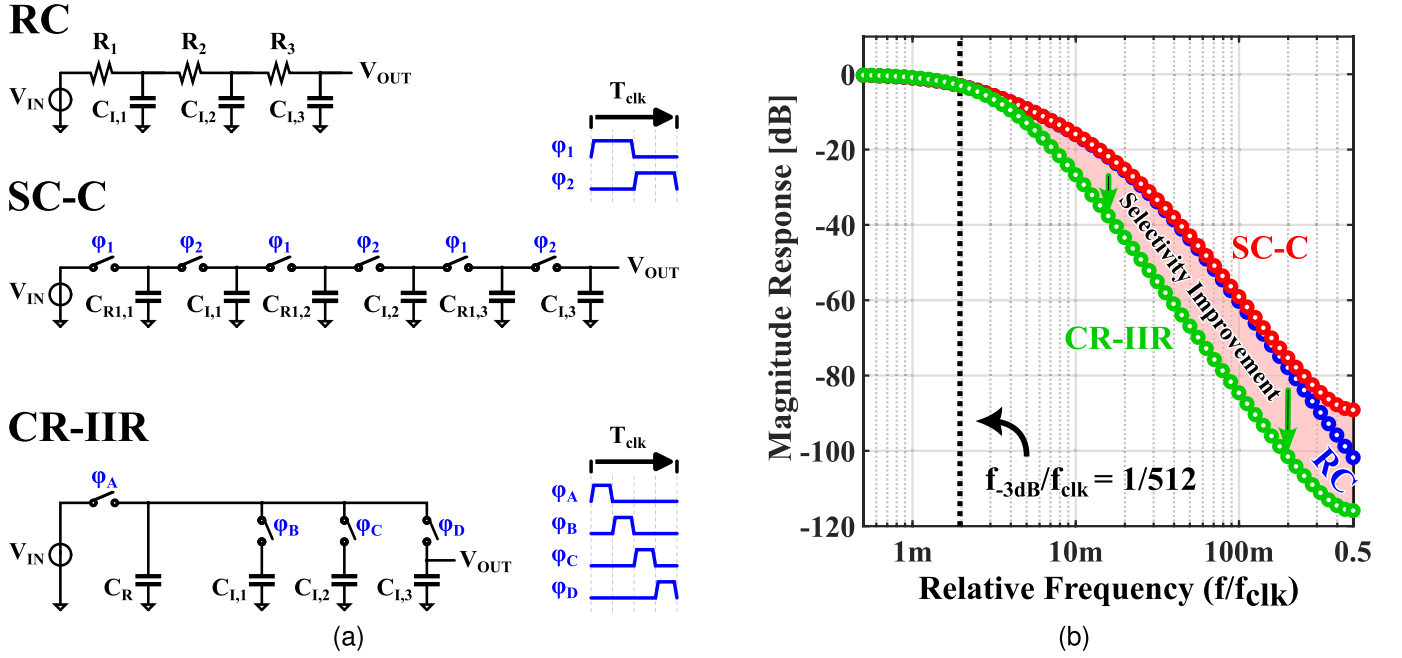


Fig. 2. Comparison of (a) third-order Resistor-Capacitor (RC), Switched-Capacitor-Capacitor (SC-C) and Charge-Rotating IIR (CR-IIR) filters and (b) their associated (sampled PAC) simulated magnitude responses for equal-size integration capacitors ($C_{i,1} = C_{i,2} = C_{i,3}$), equal size resistors ($R_1 = R_2 = R_3$), equal size SC-C switching capacitors ($C_{R1,1} = C_{R1,3} = C_{R1,3}$) and equal -3 dB bandwidths (achieved by setting $C_R \neq C_{R1,1}$). The CR-IIR filter exhibits significantly higher rejection beyond the -3 dB frequency than the SC-C (and RC) filter(s), as indicated by the translucent red area.

A. RC and SC-C Cascade Filters

When cascading RC or Switched-Capacitor-C (SC-C) stages (as illustrated in Fig. 2a), the frequency-dependent input impedance of each added stage loads the previous sections. For the RC filter, this can be understood by noting that the voltage on $C_{i,1}$ is influenced not only by the input voltage V_{in} through resistor R_1 , but also by the voltage on $C_{i,2}$ through resistor R_2 , which is a low-pass filtered version of V_{in} . As a result, $C_{i,1}$ experiences frequency-dependent loading. In SC-C cascade filters, a similar effect occurs where the voltage on $C_{i,1}$ is not only influenced (during ϕ_2) by the input voltage V_{in} (sampled onto $C_{R1,1}$ at the end of ϕ_1), but also (during ϕ_1) by the voltage on $C_{R1,2}$ (a sample of the voltage on $C_{i,2}$ from the end of ϕ_2). As a result, the (magnitude) response of these filters exhibits a shallow initial roll-off beyond their -3 dB frequency f_{-3dB} , thus yielding poor out-of-band frequency selectivity as can be seen in Fig. 2b. Consequently, the use of such filters in $\Delta\Sigma$ loops yields poor noise-shaping performance [16], as indicated by (1).

B. The CR-IIR Filter

Because the inter-stage loading effect in SC-C cascade filters only occurs in ϕ_1 between sections, eliminating this phase would avoid the effect. The fully-passive voltage-sampling variant of the active Charge-Rotating IIR (CR-IIR) filter [17] achieves this by utilizing a single switching capacitor (C_R in Fig. 2a), which samples the input voltage V_{in} at the end of ϕ_A . After sampling the input, C_R consecutively charge-shares with integration capacitors $C_{i,1}$, $C_{i,2}$ and $C_{i,3}$. The (history) charge on C_R , carried over from the previous clock cycle, is dissipated during ϕ_A . Therefore, the voltage on $C_{i,1}$ is only influenced by V_{in} , and not by the voltages on $C_{i,2}$ and $C_{i,3}$, and the inter-stage loading effect is eliminated. Additionally, the separate reset phase, implemented in [17] as

required by its active CR-IIR filter mode, is not needed here. The resulting improved frequency selectivity that this filter offers, illustrated in Fig. 2a, grants an increased quantizer gain as per (1) and thereby improved noise-shaping. This makes it a strong candidate for use in $P\Delta\Sigma$ Ms, and it is therefore applied in this work. Finally, we note a further benefit of the CR-IIR filter: its DT nature renders its transfer function readily flexible with clock frequency.

V. CR-IIR LOOP FILTER DESIGN FOR $P\Delta\Sigma$ MS

After establishing the filter architecture around which we will construct our $P\Delta\Sigma$ M loop filter, we must determine its component values. To do this, we begin by determining the pole location(s) as a function of the $P\Delta\Sigma$ M bandwidth $BW_{\Delta\Sigma} = f_{clk}/(2 \cdot OSR)$, which we use as a primary design variable, followed by the noise level of the filter, and give design equations for both.

Dimensioning the loop filter poles such that its cut-off frequency f_{-3dB} is far below the modulator bandwidth $BW_{\Delta\Sigma}$ (as would be the case in a typical active $\Delta\Sigma$ M utilizing active integrators with high DC gain) results in a high quantizer gain A_Q as per (1). However, it also incurs significant in-band attenuation in $H_{FB}(f)$, thus increasing demands on the thermal noise performance of the quantizer ADC. Contrarily, setting the poles such that $f_{-3dB} \gg BW_{\Delta\Sigma}$ yields a $H_{FB}(f)$ with poor rejection of out-of-band quantization noise (including at $f_s/2$), resulting in low quantizer gain and thus poor noise-shaping performance. A practical compromise is to set $f_{-3dB} = BW_{\Delta\Sigma}$, as is done in this work.

In SC filters, bandwidth (relative to the clock frequency) is set by the ratio between the switching- and integrating capacitors [17] (C_R and C_I , respectively, from Fig. 2a). Under the established filter bandwidth constraint ($f_{-3dB} = BW_{\Delta\Sigma}$), this capacitor ratio is linked to the OSR of the $P\Delta\Sigma$ M in

which the filter is applied. Unequal filter poles (integration capacitors) result in a shallow initial magnitude response roll-off beyond $f_{-3\text{dB}}$, similar to the effect caused by inter-stage loading. Coinciding the loop filter poles grants maximal out-of-band rejection and is thus the logical choice for a $\Delta\Sigma\text{M}$ as per (1). The transfer function of such an N^{th} -order coincident-pole CR-IIR filter is given in [17] as

$$\frac{V_{\text{out}}(z)}{V_{\text{in}}(z)} = z^{-\left(\frac{N}{N+1}\right)} \cdot \left(\frac{1}{1 - \frac{(C_I/C_R)+1}{(C_I/C_R)+1} \cdot z^{-1}} \right)^N \quad (6)$$

which can be used to derive the ratio between the (N equal-size) integration capacitor(s) C_I and switching capacitor required to achieve $f_{-3\text{dB}} = f_{\text{clk}}/(2 \cdot \text{OSR})$ as

$$\frac{C_I}{C_R} = \sqrt{\frac{1}{4} + \frac{\sqrt{2}-1}{2-2\cos\left(\frac{\pi}{\text{OSR}}\right)} - \frac{1}{2}}. \quad (7)$$

After constraining the ratio between C_R and C_I via the OSR, we must determine the noise (and thus impedance) level of the filter. Using (3.37) from [18], it can be shown that the integrated in-band, sampled output-referred noise contribution of an arbitrary-order charge-rotating IIR filter with $f_{-3\text{dB}} = f_{\text{clk}}/(2 \cdot \text{OSR})$, can be written as

$$\overline{v_{n,\text{out},\text{in-band}}^2} \approx \frac{2k_B T}{C_R \cdot \text{OSR}} \quad (8)$$

where k_B is Boltzmann's constant, T is the absolute temperature in Kelvin. The $2k_B T/C_R$ term accounts for the $k_B T/C$ noise (samples) produced in the sampling (first) phase and last charge-sharing phase, with the noise contribution from the other charge-sharing phases being negligible [18]. The factor $1/\text{OSR}$ represents the fraction of this total $k_B T/C$ noise that is present in-band.

These two equations provide a basis for our $\text{P}\Delta\Sigma\text{M}$ design approach. A discussion on switch design follows in Section VI-D.

VI. CIRCUIT IMPLEMENTATION

This section discusses the circuit-level implementation of the passive third-order 1 bit $\Delta\Sigma\text{M}$ implemented in this work. Although we have utilized GlobalFoundries' 22 nm FD-SOI process, our design techniques are also applicable to standard CMOS technologies. A simplified schematic representation of the positive half of the proposed differential design, along with its capacitor values, is shown in Fig. 3a. Its timing diagram is shown in Fig. 3b. DAC1 forms the primary feedback path, with DAC2 and DAC3 serving to stabilize the modulator, as will be discussed later. Equal-size capacitors $C_{R,\text{IN}}$ and $C_{R,\text{DAC}}$ sample the input and DAC voltage, respectively. They combine in parallel from the first-charge-sharing phase onward, to simultaneously achieve feedback summation and charge-sharing with integration capacitor $C_{I,1}$. Afterwards, they consecutively charge-share with integration capacitors $C_{I,2}$ and $C_{I,3}$, resulting in a loop filter exhibiting a third-order roll-off. Numerical simulations (without mismatch) indicated that achieving the performance targets of 80 dB SNDR and 4 MHz bandwidth required a third-order loop filter and OSR of $256\times$. The resulting NTF, STF, and loop filter responses $H_{\text{FF}}(f)$ and $H_{\text{FB}}(f)$ are shown in Fig. 4. Initially assuming

a Maximum Stable Amplitude (MSA) equal to the DAC full-scale and white quantization noise, the 60 dB in-band NTF suppression and OSR grant a SNQ_R of more than 91 dB. Even with a slightly lower practical MSA, the margin between the SNQ_R and SNDR is well over 6 dB. Because the loop filter and comparator (thermal) noise contribute similarly to overall $\text{P}\Delta\Sigma\text{M}$ input noise, the thermal noise budget was split equally among them.

A. Loop Filter

The previously mentioned equal-size capacitors $C_{R,\text{DAC}}$ and $C_{R,\text{IN}}$ in Fig. 3 together make up the switching capacitor C_R , with their combined value of 560 fF chosen using (8). An inversion in the 1 bit DAC signal ensures negative feedback. Four slices each containing one $C_{R,\text{DAC}}$ and one $C_{R,\text{IN}}$ are time-interleaved, allowing each individual slice to utilize a larger portion of T_{clk} for input signal tracking and sampling, thereby relaxing the challenge of achieving high-linearity. Thanks to this time-interleaving, the filter performs the sampling and all charge-sharing clock subphases during each T_{clk} , analogous to the CR-IIR circuit in Fig. 2a. As such, the output-referred result of (8) still holds for our loop filter. However, the (approximately) 6 dB attenuation in both $H_{\text{FF}}(f)$ and $H_{\text{FB}}(f)$ results in an equally increased input-referred contribution from both the loop filter noise and the quantizer ADC noise. Additionally, mismatch between the interleaved switching capacitor slices introduces time-variance, which frequency-translates quantization noise around $f_{\text{clk}}/2$ and $f_{\text{clk}}/4$ in-band, resulting in an increased noise floor. This effect is discussed in Section VI-F.

As can be seen in Fig. 3a, the comparator produces a 1 bit decision from the loop filter output voltage, at the top-plates of integration capacitors $C_{I,3}$, during the time that the latter are not charge-sharing with $C_{R,\text{DAC}}$ and $C_{R,\text{IN}}$. This ensures that only settled loop filter output voltage samples are processed by the $\Delta\Sigma$ loop.

To understand the purpose of each DAC in Fig. 3, let us first consider the transfer function towards the loop filter output (quantizer input) of each individual DAC. The global feedback path via DAC1, through its associated switching capacitor $C_{R,\text{DAC}}$, exhibits a third-order DT frequency response (similar to $H_{\text{FF}}(f)$ in Fig. 4). Moreover, the DAC1 sample from phase φ_A only affects the loop filter output voltage in phase φ_D , three clock cycles later. Together, this results in a large phase lag near $f_{\text{clk}}/2$, which would destabilize the $\Delta\Sigma\text{M}$ if not properly compensated. The local feedback path via DAC3, through its associated switching capacitor $C_{Z,2}$, couples directly into the top-plates of $C_{I,3}$ within one clock cycle, granting it a first-order magnitude response and low phase lag. This is analogous to coupling a switched-capacitor DAC (in DT $\Delta\Sigma\text{Ms}$) or a resistor or current-steering DAC (in CT $\Delta\Sigma\text{Ms}$) into the last integrator in Active $\Delta\Sigma\text{Ms}$. The DAC1 and DAC3 paths combine at the top-plates of $C_{I,3}$, and the ratio of $C_{Z,2}$ to $C_{R,\text{DAC}}$ sets the frequency at which the response of $H_{\text{FB}}(f)$ transitions from third- to first-order. However, because the difference in phase lag between the two paths is approximately 180° , a finite anti-resonance occurs around this transition frequency. The absence of this anti-resonance in the response of $H_{\text{FF}}(f)$ causes the STF of the $\Delta\Sigma\text{M}$ to exhibit a resonance (peaking). To reduce this, the quasi-local

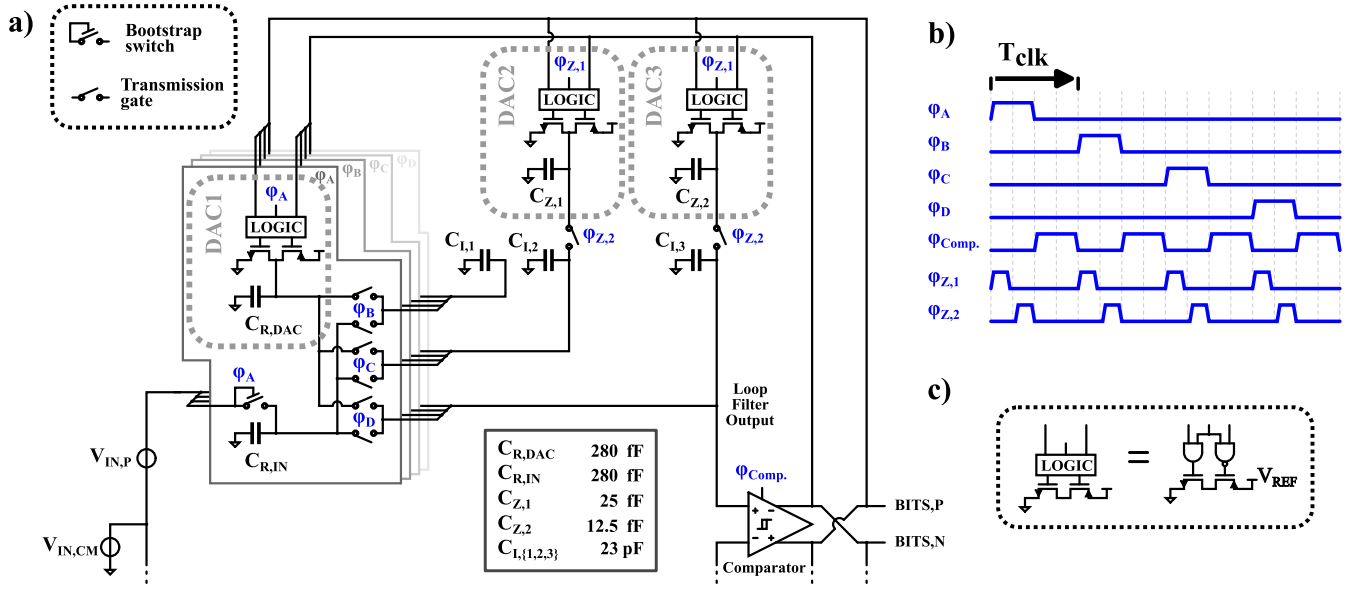


Fig. 3. a) Simplified schematic of the positive half of the pseudo-differential loop filter with all capacitor values, b) timing diagram and c) tristate inverter DAC implementation. The DAC (reference) supply voltage V_{REF} is provided off-chip.

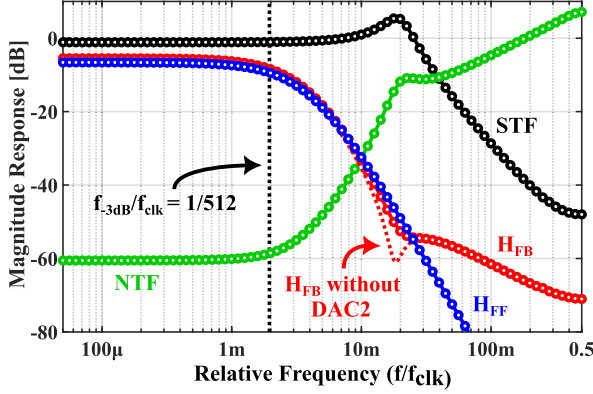


Fig. 4. Simulated (linearized, sampled PAC) NTF, STF, $H_{FF}(f)$ and $H_{FB}(f)$ of the implemented Passive $\Delta\Sigma$ M with OSR = 256 without mismatch-induced degradation due to time-interleaving in the loop filter.

feedback path via DAC2, through its associated switching capacitor $C_{Z,1}$, couples into top-plates of $C_{1,2}$ (analogous to a DAC coupling into the second-last integrator in Active $\Delta\Sigma$ Ms), thus exhibiting a second-order response. As a result, the combined phase response of DAC1 and DAC2 reduces to second-order before the transition frequency, thereby damping the anti-resonance. The result is a Cascade of Integrators with FeedBack (CIFB) type compensation [4].

Achieving large in-band NTF suppression requires high quantizer gain, which result from maximizing the high-frequency rejection of $H_{FB}(f)$ as indicated by (1). While the removal of DAC2 and DAC3 would achieve this, the resulting modulator would be unstable. Thus, there must exist some (near-)optimal transition frequency, with associated coupling strength ratio between DAC1 and DAC3, which maximizes rejection while simultaneously avoiding instability. In active $\Delta\Sigma$ M, this coupling ratio can be achieved by applying voltage gain in the first and second integrators. However, in our proposed passive $\Delta\Sigma$ M, this is instead achieved by the large ratio between $C_{R,DAC}$ and $C_{Z,2}$ given in Fig. 3. The values of $C_{Z,1}$ and $C_{Z,2}$ chosen to achieve the desired CIFB

compensation for the 3rd-order CR-IIR filter (shown in Fig. 3) were found through extensive numerical simulations.

In contrast to our approach, which only uses two small switching capacitors ($C_{Z,1}$ and $C_{Z,2}$) to achieve CIFB compensation, the 2nd-order P $\Delta\Sigma$ M in [16] contains an additional large capacitor, equal in size to the integration capacitors, which introduces a zero in the loop filter response for stability. Decreasing the size of this large capacitor would shift this zero to a lower frequency, reducing the high-frequency rejection of the loop filter and thereby the (in-band) quantization noise suppression of the P $\Delta\Sigma$ M in [16], as per our (1).

B. 1 bit Quantizer ADC

Because the quantizer is 1 bit, a single comparator is sufficient to implement its ADC, for which a modified version of the high-speed dynamic design from [14], shown here in Fig. 5, is utilized. Capacitive neutralization of the latch stage using capacitors $C_{Neut.}$, along with a tail charge pump which increases the overdrive of the input pair (M1 and M2), enhances its speed over previous dynamic comparator designs [13]. The negative excursion of the voltage at the top plate of C_{tail} is limited to 160 mV. Because the preamplifier gain is approximately 10 $\times$, the (input-referred) noise contribution of the latch is minimized. As a result of the charge pump, its input pair MOSTs M1 and M2 operate in strong inversion to achieve high speed operation. Despite this, the comparator's noise-power trade-off is similar to that given by (5), which was derived for weak inversion, due to its similar dynamic architecture.

Because the original comparator design from [14] was limited to operation at 1 GHz, modifications were made to increase its speed and enable its use in the 2.048 GHz P $\Delta\Sigma$ M in this work. Specifically, equations (3) and (5) from Section III-B indicate that achieving a given comparator input noise level only requires the consumption of a given charge packet from the power supply, but is independent of preamplifier integration time. Thus, we maximized preamplifier

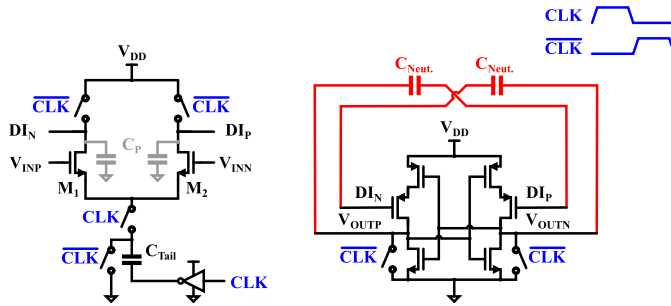


Fig. 5. Simplified schematic of the (modified) high-speed dynamic comparator from [14] used this work. Neutralization capacitors C_{Neut} , used to enhance comparator speed are highlighted red. Preamplifier integration capacitors C_P (grey) consist purely of MOST and layout parasitics to maximize speed.

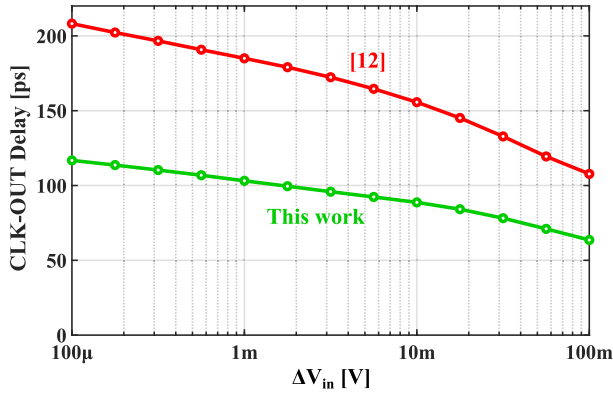


Fig. 6. Simulated Clock-to-Out (CLK-OUT) delay versus differential input voltage ΔV_{in} of both the comparator from [14] at 0.8 V and the modified design used here at 0.9 V, showing a nearly halved CLK-OUT delay for small input signals.

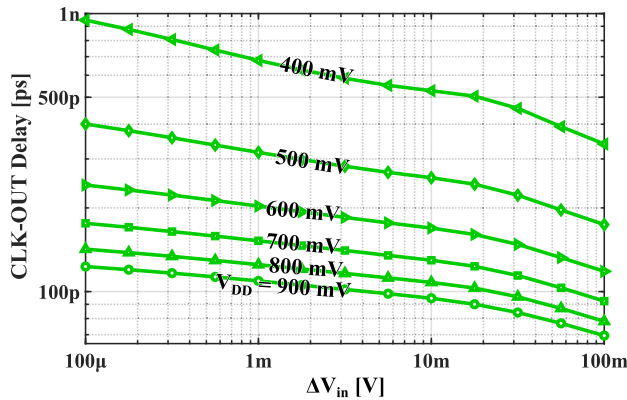


Fig. 7. Simulated Clock-to-Out (CLK-OUT) delay versus differential input voltage ΔV_{in} the proposed comparator (modified from [14]) at 0.9 V and several (lowered) supply voltages of 0.8, 0.7, 0.6, 0.5 and 0.4 V. The input common-mode voltage for each curve is $V_{\text{DD}}/2$.

speed in this work by modifying its integration capacitors C_P to consist purely of (MOST and layout) parasitics. This maximizes transistor width for a given (parasitic) C_P , allowing M1 and M2 to rapidly conduct the charge needed to obtain the input-referred voltage noise of $120\mu\text{V}$ required in our modulator. Additionally, its supply voltage was raised from 0.8 V to 0.9 V. Together these measures enable it to achieve the required speed of operation, which is illustrated by comparison of its simulated Clock-Out (CLK-OUT) delay versus differential input voltage ΔV_{in} to that of the design from [14] in Fig. 6.

To verify the supply-scalability of this comparator design, we also simulated the CLK-OUT delay at reduced supply voltages of 0.8, 0.7, 0.6, 0.5 and 0.4 V, with the input common-mode voltage set to $V_{\text{DD}}/2$. The resulting CLK-OUT delay versus ΔV_{in} is shown in Fig. 7.

C. 1 bit Quantizer DAC

As the loop filter is switched-capacitor-based, switched-capacitor DACs with their jitter insensitivity are a natural choice [4]. As shown schematically in Fig. 3c, they are implemented as tri-state inverters, which sample a 1 bit voltage on switching capacitor $C_{\text{R,DAC}}$, $C_{\text{Z},1}$ or $C_{\text{Z},2}$. Clock-driven AND and NAND gates ensure that the DAC inverters are in their high-Z state (i.e. both n- and pMOST are off) when their associated switching capacitor ($C_{\text{R,DAC}}$, $C_{\text{Z},1}$ or $C_{\text{Z},2}$) charge-shares with an integration capacitor. This is critical, as otherwise the latter would be shorted to one of the DAC inverter supplies, destroying loop filter selectivity.

D. Switch Implementation

Due to the fact that DAC1 couples into the loop filter much more strongly than DAC2 and DAC3, its output contains a near-complement of the input signal. Therefore, signal swing reduces substantially, approximately 20 dB, after $C_{\text{R,DAC}}$ and $C_{\text{R,IN}}$ charge-share with $C_{\text{I},1}$. Further reduction occurs when DAC2 and DAC3 couple into $C_{\text{I},2}$ and $C_{\text{I},3}$, respectively. This relaxes linearity requirements on later charge-sharing switches, allowing the second be smaller than the first, and the third smaller still, reducing clock power. The charge-sharing switches are implemented as transmission gates.

Contrary to the charge-sharing switches, the input sample switch must sample the full-scale continuous-time input signal with sufficiently high ($< -80\text{ dB THD}$) linearity in the allotted $T_{\text{clk}}/2$ sample window. Because the on-resistance (R_{on}) of a MOST switch is non-linear, significant distortion is produced unless the sampler settles to a given degree before the input sample is taken by opening the switch [20]. With increasing clock frequency, an increasingly wider (i.e. lower R_{on}) MOST switch is required to achieve the same degree of settling in the decreasing sample time window. Unfortunately, this increases the magnitude of the (non-linear) channel charge injected into the sampling capacitor when the MOST switch is opened [20]. As a result, there exists a minimum combined distortion power from both effects for a given MOST switch width for each distinct clock frequency. Additionally, this minimum distortion power increases with clock frequency, thereby limiting the maximum possible speed of operation for a given linearity demand. However, the minimum distortion power can still be improved by making both switch R_{on} and (non-linear) charge-injection less dependent on the input signal by standard bootstrap techniques [20] such as the one from [21] applied in this work. Still, the increasing minimum distortion power with increasing clock frequency produced by the input sample switch presented a bottleneck to the maximum speed of operation of our design.

E. Loop Filter Clock Phase Generation

There are several key requirements on the $\Delta\Sigma$ clock phases. Firstly, it is of critical importance that there exists

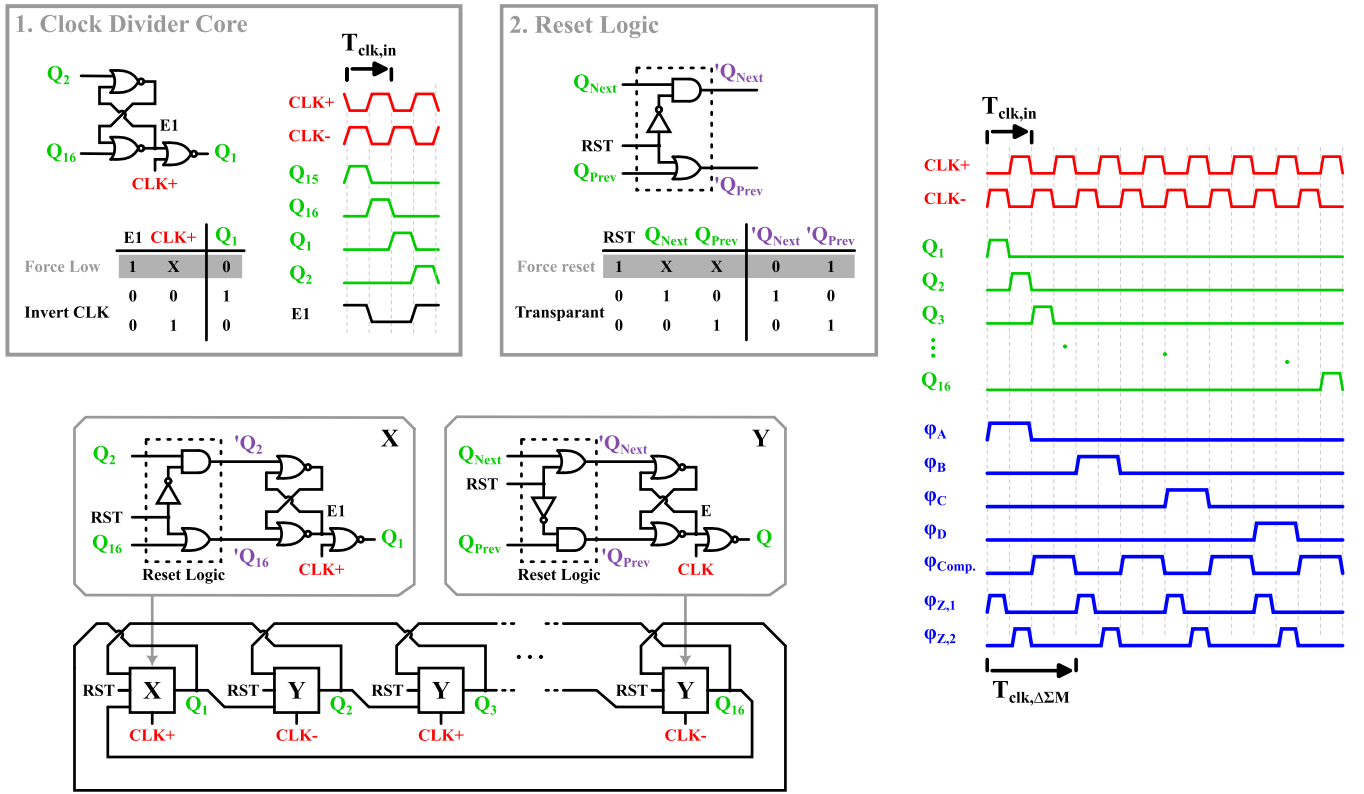


Fig. 8. Simplified schematic diagram of the timing generation circuitry. The 16-phase clock divider (based on [19]) utilizes both edges of the differential input clock (CLK+, CLK-). Reset logic allows presetting of the clock divider states to ensure that only a single output phase of the clock divider is high at any given time. Recombination and non-overlap logic (not shown) recombines the 16 phases from the clock divider into the non-overlapping clock phases from Fig. 3.

no overlap between the clock waveforms of the sampling- and charge-sharing switches. Secondly, to ensure predictable DT behavior, the comparator should only operate after filter charge-sharing has settled. Finally, DAC3 should couple into the loop filter during each clock cycle, to avoid excess loop delay. The timing generation circuitry shown in Fig. 8 was designed to meet these requirements.

A 16-phase clock divider, based on [19] for its ability to utilize both edges of the incoming differential clock, was implemented. Its outputs are combined into the needed $P\Delta\Sigma M$ clock phases using conventional recombination and non-overlap logic.

To understand its operation, let us assume that the divider is working correctly and only a single output pulse ($Q_1 \dots Q_{16}$) is high at any given time. We first consider the clock divider core and its truth table illustrated in Inset 1. of Fig. 8. A NOR gate receiving a CLK and E(nable) signal produces an output Q, which equals the inverted CLK signal when Enable is low, being always low otherwise. This Enable signal is generated by a NOR latch. By using the previous divider output phase to set Enable low, and the next divider output phase to set it high again, only a single CLK pulse is passed to the divider output Q, as illustrated in the timing diagram.

Although we assumed that only a single output phase Q was active at any time, this may not be the case if the NOR latches are not properly initialized. This would allow more than one divider output phase to be high at a given time, disrupting proper operation of the $P\Delta\Sigma M$ loop filter. To force correct initialization, the reset logic shown in Inset 2. of Fig. 8 was

added to each clock divider core block's NOR latch. When RST is high, the reset logic forces the NOR latch in the X unit block into the enable low state (i.e. invert CLK), regardless of the state of its two divider phase inputs Q. By the same principle, but flipping the control logic with respect to unit block X, all NOR latches in the Y unit blocks are forced into their enable high state (i.e. force low). When RST is returned low, the reset logic becomes transparent from its Q inputs to its 'Q outputs, and clock divider operation continues as if the reset logic was not present.

F. SNR Degradation Due to Mismatch-Induced Interleaving Mixing

Capacitor mismatch between the four interleaved switched-capacitor slices in Fig. 3a, each containing a $C_{R,DAC}$ and a $C_{R,IN}$, results in Linear Periodically Time-Variant (LPTV) behavior with a periodicity of $4 \times T_{clk}$ in the step response of the loop filter. As a result, a fraction (related to the degree of mismatch) of the quantization noise in the DAC signal around $f_{clk}/4$ and $f_{clk}/2$ frequency-translates back into the modulator bandwidth [22]. Due to the noise-shaping action of the $\Delta\Sigma M$, there is significantly more noise power around these frequencies than there is in-band. Thus, even a small fraction of this noise power frequency-translating down leads to a significant SN(D)R degradation, resulting in a high sensitivity to mismatch. The incurred performance degradations are present in the measurements discussed in the following section.

VII. MEASUREMENT RESULTS

Fig. 9 shows a die photograph of the passive $\Delta\Sigma$ implemented in GlobalFoundries 22 nm FD-SOI CMOS.

The packaged passive $\Delta\Sigma$ sample is mounted in an Ironwood ZIF socket on a test PCB. Clock and input signals are provided by an R&S SGS100A and two Keysight 33600A generators, respectively. Differential output bit data is captured using a Tektronix MSO64B oscilloscope. The bit data is loaded into MATLAB, where it is re-timed and re-quantized, after which the first 2^{17} bits are Hann-windowed and processed using an FFT. To obtain reduced bin variance, the shown FFT spectra are the result of magnitude-averaging 10 FFTs, each produced from consecutively captured bitstreams. All reported measurements were taken at $f_{\text{clk}} = 2.048$ GHz, unless otherwise mentioned, resulting in an FFT Resolution BandWidth (RBW) of 15.625 kHz. The OSR of $256\times$ yields a 4 MHz modulator bandwidth. Fig. 10 shows the peak SNDR and peak SNR for all nine measured samples, exhibiting spread originating from the matching-sensitive folding of spectral content around $f_{\text{clk}}/2$ and $f_{\text{clk}}/4$. All remaining shown measurements were performed on sample #4, with its roughly average performance among all measured samples. We intentionally did not select the sample with the highest SN(D)R.

Fig. 11 shows the output spectrum of the $\Delta\Sigma$ for a 0 dBFS input tone placed one RBW below one-third of the modulator's 4 MHz bandwidth, so that the upper bin of the HD3 is at the band edge, in blue. The resulting HD2 and HD3 magnitudes are -81.3 dBFS and -77.1 dBFS, respectively. These measured linearity results exhibit a reasonable match to numerical Monte Carlo simulations which included mismatch between the time-interleaved CR-IIR loop filter slices, suggesting that this mismatch does not have a significant impact on linearity. This is reasonable in light of Section VI-F, where the discussed impact of mismatch among the time-interleaved CR-IIR loop filter slices is in principle LPTV. The tone at $f_{\text{clk}}/4$, related to the interleaved operation of the loop filter could be easily removed by a (sinc) notch in the frequency response of the digital decimation filter. The NTF from Fig. 4 can be recognized in the shape of the noise floor. The measured SNDR and SNR versus input signal level is shown in Fig. 12, reaching a maximum of 71.9 dB and 72.6 dB, respectively. Additionally, an idle measurement (zero differential input signal applied) showed no visible idle tones. The presented one-tone measurement does not give an adequate indication of modulator linearity at full bandwidth, as it was performed with an input signal slightly below BW/3 to ensure an in-band HD3. Therefore, a two-tone measurement with two -6 dBFS tones, the first at the modulator bandwidth of 4 MHz and the second 1 RBW above 4.5 MHz, respectively, was carried out. The resulting magnitude-averaged spectrum in Fig. 13 shows the two -6 dBFS input tones and resulting IM2, IM3 and IM4 products at -80.8 dBFS -83 dBFS and -88.8 dBFS, respectively. This indicates that the modulator maintains good linearity, even at the band-edge.

To investigate the intermodulation behavior of the implemented modulator, we performed a far out-of-band two-tone measurement test. Such a situation can be found in e.g. radio applications, where large far out-of-band interfering signals may be present. The two -6 dBFS input tones were placed

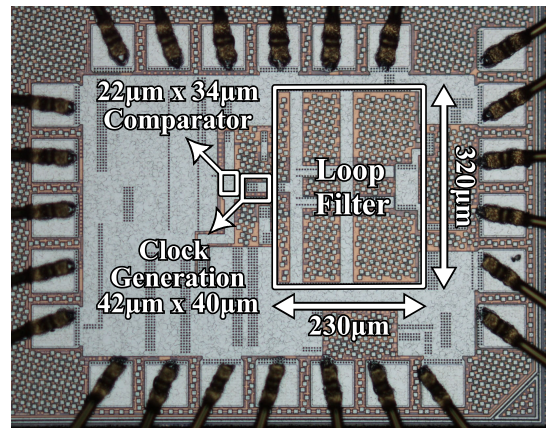


Fig. 9. Die photograph of the prototype Passive $\Delta\Sigma$ implemented in GlobalFoundries 22 nm FD-SOI technology.

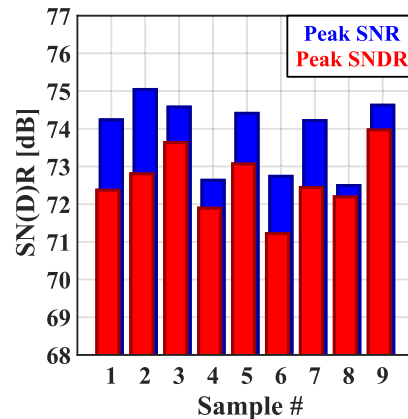


Fig. 10. Peak SNDR and peak SNR for all nine measured samples. Intermediate-performing sample #4 was selected for all further measurements.

at 40 MHz (i.e. $10\times$ BW) and at one RBW above 76.5 MHz, ensuring that the resulting IM3 product ends up in-band and at the same frequency as for the previous band-edge two-tone test. The measured spectrum along with the STF of is shown in Fig. 14. Although one of the input tones was placed at the STF resonance frequency, where the loop is most sensitive to overload, the resulting IM3 product is barely visible above the noise floor at -92.4 dBFS. Even a trivial (passive) first-order filter with its cut-off at the modulator band-edge would reduce the lowest of the two interfering input tones by nearly 20 dB.

The prototype modulator consumes 4.43 mW at 4 MHz bandwidth, of which 3.15 mW (71.1%) is clock power, 0.94 mW (21.2%) is comparator power (including local clock buffer power) and 0.34 mW (7.7%) is DAC power. Consumption of the clock divider and non-overlap recombination logic are included in the reported clock power. Comparator power consumption is higher than predicted by the fundamental limit given in Section III-B, due to optimization of its design for high speed operation.

We can estimate the impact of input common-mode voltage variations by noting that when the input common-mode voltage is raised or lowered away from $V_{\text{DD}}/2$ by some amount, a signal amplitude reduction of equal magnitude is required to avoid clipping the modulator. This limitation arises due to the finite DAC full-scale, and is present in all ADC designs. An additional performance degradation can arise due to an imbalance in the differential input signals presented to the $\Delta\Sigma$, resulting in an input common-mode

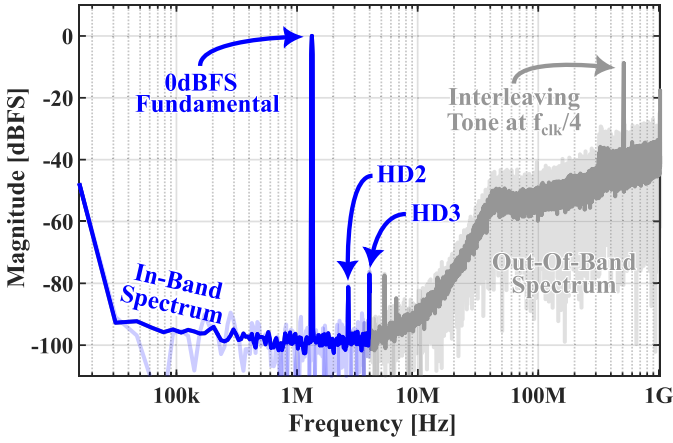


Fig. 11. Measured one-tone FFT averaged magnitude spectrum with one 0dBFS input tone band placed one RBW below BW/3 so all three HD3 bins fall inside the 4MHz bandwidth, showing a -81.3 dBFS HD2 and -77.1 dBFS HD3. The translucent spectrum is one of the ten raw FFT magnitude spectra used to produce the $10\times$ averaged magnitude spectrum.

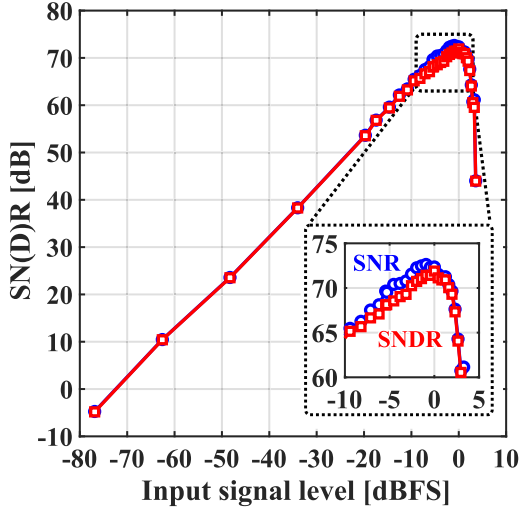


Fig. 12. Measured SNDR and SNR versus input signal magnitude using one input tone at a frequency slightly below BW/3 as in Fig. 11, showing a peak SNDR of 71.9 dB and a peak SNR of 72.6 dB.

TABLE I

MEASURED SNDR, POWER CONSUMPTION AND BANDWIDTH AT SEVERAL (LOWERED) SUPPLY VOLTAGES. THE SAME LOWERED V_{DD} IS APPLIED TO ALL $\Delta\Sigma$ CORE SUPPLIES: CLOCK, COMPARATOR AND DAC

V_{DD} [V]	0.4	0.5	0.6	0.7	0.8	0.9
Bandwidth [kHz]	156.25	625	1250	1250	2500	4000
Power [μ W]	37.8	208.8	604.7	848.7	2245	4430
Peak SNDR [dB]	62.0	65.6	66.3	67.5	71.2	71.9
FoM _{Schreier} [dB]	158.2	160.4	159.4	159.2	161.6	161.7

$$* \text{FoM}_{\text{Schreier}} = \text{SNDR}_{\text{peak}} + 10 \log_{10} \left(\frac{\text{BW}}{P} \right).$$

voltage variation at the signal frequency. The presence of this AC common-mode voltage reduces the effective full-scale by the mechanism described above. Additionally, it also causes the AC signals in each pseudo-differential half of the circuit to be unequal in amplitude. As a result, even-order distortion terms with unequal amplitudes are produced in

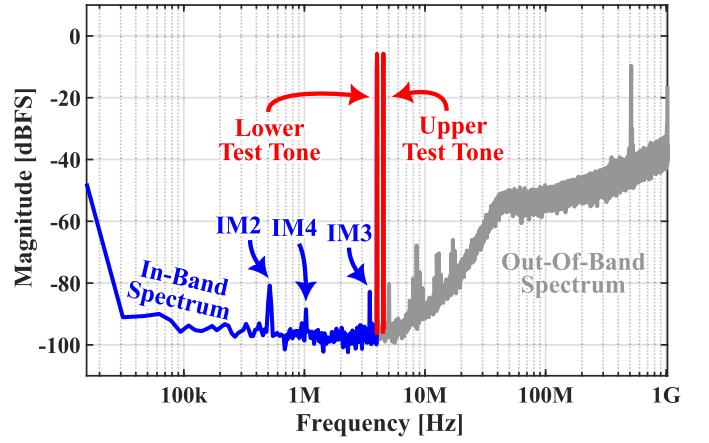


Fig. 13. Measured magnitude spectrum with two -6 dBFS input test tones, showing a -80.8 dBFS IM2, -83 dBFS IM3 and -88.8 dBFS IM4. The lower test tone is at the band-edge (4 MHz) and the upper test tone (1 RBW above 4.5 MHz) is placed to ensure that both the IM2 and IM3 products fall in-band at distinct frequencies.

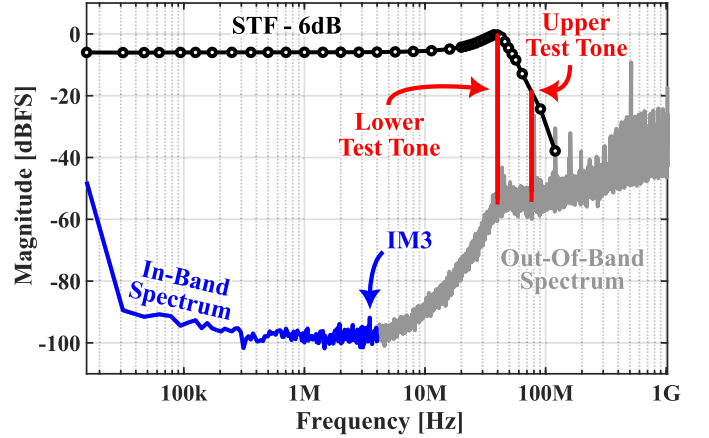


Fig. 14. Measured magnitude spectrum with two -6 dBFS input tones, showing an IM3 product which is barely visible above the noise floor at -92.4 dBFS. The lower tone is at 10 times the modulator bandwidth (40 MHz) and the upper tone (1 RBW above 76.5 MHz) is placed to ensure the IM3 product falls in-band. The modulator's measured STF, reduced by 6 dB, is shown to indicate the expected output signal magnitude due to the -6 dBFS test tones.

the two halves of the circuit. This creates differential even-order distortion terms, which may degrade the linearity of the $\Delta\Sigma$. Numerical simulations without mismatch indicate that an AC input common-mode voltage signal does indeed reduce the effective full-scale of our $\Delta\Sigma$ and thereby its peak SNDR. However, even a 200 mVpkpk common-mode voltage (with an input window of 1.6 Vpkpk) at the signal frequency, slightly below BW/3, does not impact the SNDR by more than 0.1 dB for (reduced) signals within the (reduced) full-scale. Further Monte Carlo simulations including mismatch, with the same 200 mVpkpk common-mode voltage signal and associated reduced full-scale, show that further degradation of the peak SNDR remain within 1 dB, even in the face of mismatch.

A key advantage of $P\Delta\Sigma$ s over their active counterparts is the absence of (highly) supply voltage-sensitive linear amplifiers, allowing them to continue operating even at significantly lowered supply voltages, albeit at reduced bandwidth (speed). The DT nature of our $P\Delta\Sigma$ readily

TABLE II
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART SINGLE-LOOP, 1 BIT PASSIVE $\Delta\Sigma$ MS AND SIMILAR-PERFORMING SINGLE-LOOP, 1 BIT ACTIVE $\Delta\Sigma$ MS MEASURED IN LITERATURE

	PASSIVE										ACTIVE		
	This work		Wang [16]	Chen [9] [6]**		Yeknami [8]**	Somappa [28]** [29]**		ElAnsary [30] [31]		Jiang [23]	Baluni [24]	Wang [25]
CMOS Technology [nm]	22 FD-SOI		28	1200	130	65	180	180	130	130	28	65	180
Oversample Ratio	256		256	250	520	500	128	128	100	100	208	64	80
Modulator Order	3 rd		2 nd	2 nd	2 nd	2 nd	2 nd	2 nd	1 st	2 nd	2 nd	4 th	3 rd
Discrete/Continuous Time	DT		DT	DT	DT	DT	CT	CT	DT	DT	DT	CT	CT
V _{DD} [V]	0.9	0.4	N/A	3.3	1.5	0.7	0.4	1	0.6	0.6	1/1.15/1.5	1.1	1.8
Clock Frequency [MHz]	2048	80	40.96	10	104	0.5	2.56	2.56	1	2	2080	2560	320
Bandwidth [kHz]	4000 [†]	156.25 [†]	80	20	100	0.5	10	10	5	10	5000	20000	2000
Power Consumption [μW]	4430 [†]	37.8 [†]	101.5	230	1275	0.43	0.42	0.59	0.05	0.14	23100	11300	590
SFDR [dB]	77.1 [†]	66.1 [†]	92.0	N/A	73.0	72.0	70.4	N/A	60.0	N/A	98.0	101.2 [‡]	88.4
Peak SNDR [dB]	71.9 [†]	62.0 [†]	81.1	77.0	74.1	65.0	59.1	55.4	50.3	50.0	86.1	82.1	78.7
Peak SNR [dB]	72.6 [†]	67.0 [†]	81.9	N/A	N/A	68.0	N/A	56.2	N/A	N/A	87.6	82.1	79.3
FoM _{Schreier} [*] [dB]	161.7 [†]	158.2 [†]	170.1	156.4	153.0	155.7	162.8	157.7	160.3	158.5	169.5	174.6	174.0
Active Area [mm ²]	0.076		0.059	0.400	0.123	0.125	0.046	0.068	0.004	0.010	0.070	0.37	0.13

* FoM_{Schreier} = SNDR_{peak} + 10 log₁₀ ($\frac{BW}{P}$).

** Data for highest FoM_{Schreier} entry.

[†] For an intermediate-performance sample.

[‡] After calibration.

TABLE III
PERFORMANCE COMPARISON WITH SUPPLY VOLTAGE-FLEXIBLE ADCS MEASURED IN LITERATURE

	OVERSAMPLED				NYQUIST-RATE			
	This work		Park [32]		Lee [27]		Ding [26]	
CMOS Technology [nm]	22 FD-SOI		65 LP		65		40	
Architecture	P $\Delta\Sigma$		$\Delta\Sigma$		Pipeline		SAR	
V _{DD} [V]	0.9	0.4	1	0.4	1	0.4	1.1	0.5
Clock Frequency [MHz]	2048	80	40	7.5	50	2	10	0.02
Bandwidth [kHz]	4000 [†]	156.25 [†]	400	7.5	25000	1000	5000	10
Power Consumption [μ W]	4430 [†]	37.8 [†]	948	12.7	4070	112	47.6	0.02
SFDR [dB]	77.1 [†]	66.1 [†]	N/A	N/A	85.8**	78.1**	68.8	71.3
Peak SNDR [dB]	71.9 [†]	62.0 [†]	61.1	60.5	67.7**	61.4**	58.6	54.6
Peak SNR [dB]	72.6 [†]	67.0 [†]	62.7	63.9	N/A	N/A	59.2	N/A
FoM _{Schreier} [*] [dB]	161.7 [†]	158.2 [†]	147.4	148.2	165.6	160.9	168.8	171.3
Active Area [mm ²]	0.076		0.051		0.360		0.001	

* FoM_{Schreier} = SNDR_{peak} + 10 log₁₀ ($\frac{BW}{P}$).

** After calibration.

[†] For an intermediate-performance sample.

allows the bandwidth to be reduced by lowering its clock frequency. To verify this, we measured the prototype P $\Delta\Sigma$ with all its (core) supplies (clock, comparator and DAC) at reduced voltages of 0.8, 0.7, 0.6, 0.5 and 0.4 V, while the threshold voltage of the MOSTs used in our design is approximately 250 mV. The OSR was kept constant at 256 $\times$, but each lowered V_{DD} required a reduction in clock frequency due to the resulting reduction in comparator speed. The measured maximum bandwidth was defined by the highest clock frequency beyond which the modulator either began operating at a lower limit cycle, sharply degrading performance, or ceased producing bits altogether. Table I shows the resulting maximum bandwidth, power consumption, peak SNDR and FoM_{Schreier} versus V_{DD}. Although the modulator's supply voltage was gradually reduced by a

factor 2.25, it continues to operate, albeit at reduced speed, demonstrating the supply voltage-flexibility granted by the absence of linear amplifiers. Moreover, lowering the supply voltage from 0.9 V to 0.5 V (1.8 $\times$) reduces the maximum bandwidth by only a factor 6.4. Upon lowering the supply voltage further down to 0.4 V, the bandwidth reduced by another factor 4. Numerical simulations indicate that a reduction in clock frequency (even at a fixed supply voltage) increases the impact of non-linear leakage in the first charge-sharing switch inside the CR-IIR loop filter, to the point that this effect significantly degrades overall SNDR. Because the modulator speed reduces when V_{DD} is lowered, the clock frequency must also be lowered, thereby resulting in a degradation of modulator SNDR. Additionally, as the input common-mode voltage was lowered proportionally along

with the supply voltage ($V_{CM} = V_{DD}/2$), the overdrive voltage of the comparator preamplifier MOSTs is reduced. This decreases the total charge consumption of the dynamic comparator, increasing its noise contribution (as discussed in Section VI-B), which together with the reduction in full-scale input swing due to the reduced DAC voltage, is responsible for the degradation in SNR.

Table II shows a comparison between the intermediate-performing sample #4 of our P $\Delta\Sigma$ M and other passive single-loop 1 bit $\Delta\Sigma$ Ms measured in literature. We achieve a 40 times improvement in bandwidth over the state-of-the-art while maintaining a competitive FoM_{Schreier}. While the P $\Delta\Sigma$ M in [16] also utilizes a time-interleaved CR-IIR loop filter and achieves a superior FoM_{Schreier}, it is experimentally verified at a 50 $\times$ lower (clock) frequency than our work, and only across a $\pm 20\%$ variation around their nominal supply voltage, while we have experimentally validated our P $\Delta\Sigma$ M over a range of (low) supply voltages covering 0.9, 0.8, 0.7, 0.6, 0.5 and 0.4V. We additionally compare with three recent single-loop 1 bit Active $\Delta\Sigma$ Ms with roughly similar specifications, being the DT design from [23] and the Continuous-Time (CT) designs from [24] and [25]. While these designs exhibit superior FoM_{Schreier}, they likely possess severely restricted supply voltage-flexibility due to their use of linear amplifiers, although measurements are not shown by their authors. We note that [25] is implemented in a mature and FoM-friendly 180 nm CMOS technology with a high supply voltage of 1.8 V [3]. Additionally, although Passive Noise-Shaping SAR (PNS-SAR) ADCs with excellent FoM_{Schreier} have been demonstrated in literature, their relatively low OSR [5] demands application of relatively selective anti-alias filters. As our work specifically targets avoiding the need for such selective anti-alias filters, we do not include PNS-SAR ADCs in our comparison Table II. Finally, it can be seen that P $\Delta\Sigma$ Ms have relatively large area consumption compared to active $\Delta\Sigma$ Ms. Table III benchmarks our prototype supply voltage-flexible P $\Delta\Sigma$ M against other flexible ADCs operating at or below 0.5 V in literature. While [26] and [27] operate over a similar supply voltage range at slightly superior FoM_{Schreier}, their Nyquist-rate architectures require a steep anti-alias filter which most likely demands the use of linear amplifiers. Additionally, contrary to our calibration-free design, [27] utilizes off-chip calibration to improve their SNDR by approximately 4 dB and their SFDR by over 13 dB.

VIII. CONCLUSION

In this work, we extended the bandwidth of Passive $\Delta\Sigma$ Ms (P $\Delta\Sigma$ Ms) towards 4 MHz. Two key challenges unique to the design of P $\Delta\Sigma$ Ms not found in their active counterparts were discussed. These involved the input noise criticality of the quantizer and poor selectivity of passive on-chip (loop) filters. It was shown that for the 1 bit quantizer case, the noise versus power consumption trade-off of its comparator is identical to that of the first amplifier in Active $\Delta\Sigma$ Ms. Because the absence of amplifiers disallows the use of traditional $\Delta\Sigma$ M loop filter synthesis methods, we discussed the selection of a suitable passive loop filter architecture. This lead to the selection of the voltage-sampling Charge-Rotating IIR (CR-IIR) filter, for which we derived $\Delta\Sigma$ M-oriented design equations. Measurements on the prototype 3rd-order

passive $\Delta\Sigma$ M implemented in GlobalFoundries 22 nm FD-SOI in 0.076 mm² show that it achieves a peak SNDR of 71.9 dB in a 4 MHz bandwidth while consuming 4.4 mW from a 0.9 V supply, resulting in a 161.7 dB Schreier FoM. Its 4 MHz bandwidth is a 40 times improvement over state-of-the-art Passive $\Delta\Sigma$ Ms (P $\Delta\Sigma$ Ms) in literature, which demonstrate similar Schreier FoMs. The prototype P $\Delta\Sigma$ M continues to operate as low as 0.4 V, where it achieves a reduced bandwidth of 156.25 kHz and a 62 dB peak SNDR. This allows for use of (an optimized version of) our P $\Delta\Sigma$ M in energy harvesting-powered ICs [7]. Moreover, large-signal out-of-band measurements show that the prototype modulator's in-band spectrum is not severely corrupted by such unwanted signals even a full decade beyond its bandwidth. This demonstrates rudimentary suitability for radio applications.

ACKNOWLEDGMENT

The authors would like to thank GlobalFoundries for providing silicon fabrication through the 22FDX University Program, Erwin Hardeveld and Klaas de Haan for assistance during layout, Alexander Delke and Gerard Wienk for CAD support, and Anne-Johan Annema for help in preparing the manuscript.

REFERENCES

- [1] A. -J. Annema, "Analog circuit performance and process scaling," *IEEE Trans. Circuits Syst. II, Analog Digit. Signal Process.*, vol. 46, no. 6, pp. 711–725, Jun. 1999.
- [2] B. Hershberg, S. Weaver, K. Sobue, S. Takeuchi, K. Hamashita, and U.-K. Moon, "Ring amplifiers for switched capacitor circuits," *IEEE J. Solid-State Circuits*, vol. 47, no. 12, pp. 2928–2942, Dec. 2012.
- [3] B. Nauta, "1.2 racing down the slopes of Moore's law," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2024, pp. 16–23.
- [4] S. Pavan, R. Schreier, and G. C. Temes, *Understanding Delta-Sigma Data Converters*, 2nd ed., Hoboken, NJ, USA: Wiley, 2017.
- [5] L. Jie et al., "An overview of noise-shaping SAR ADC: From fundamentals to the frontier," *IEEE Open J. Solid-State Circuits Soc.*, vol. 1, pp. 149–161, 2021.
- [6] F. Chen, S. Ramaswamy, and B. Bakkaloglu, "A 1.5 V 1 mA 80 dB passive $\Sigma\Delta$ ADC in 0.13 μ m digital CMOS process," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 1, Feb. 2003, pp. 54–477.
- [7] M. Alioto, "From less batteries to battery-less: Enabling a greener world through ultra-wide power-performance adaptation down to pWs," in *Proc. ESSCIRC*, Sep. 2022, pp. 33–40.
- [8] A. F. Yeknami, F. Qazi, and A. Alvandpour, "Low-power DT $\Delta\Sigma$ modulators using SC passive filters in 65 nm CMOS," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 61, no. 2, pp. 358–370, Feb. 2014.
- [9] F. Chen and B. Leung, "A 0.25-mW low-pass passive sigma-delta modulator with built-in mixer for a 10-MHz IF input," *IEEE J. Solid-State Circuits*, vol. 32, no. 6, pp. 774–782, Jun. 1997.
- [10] H. Chandrakumar and D. Markovic, "A 15.2-ENOB 5-kHz BW 4.5- μ W chopped CT $\Delta\Sigma$ -ADC for artifact-tolerant neural recording front ends," *IEEE J. Solid State Circuits*, vol. 53, no. 12, pp. 3470–3483, Dec. 2018.
- [11] B. Gönen, "The zoom ADC—An energy efficient ADC for high resolution," Ph.D. thesis, Dept. Electron. Instrum. Lab., Delft Univ. Technol., Delft, The Netherlands, 2021, doi: [10.4233/uuid:5bea521a-d6ac-4f6a-b45f-d2a87fec53ed](https://doi.org/10.4233/uuid:5bea521a-d6ac-4f6a-b45f-d2a87fec53ed).
- [12] M. van Elzakker, E. van Tuijl, P. Geraedts, D. Schinkel, E. A. M. Klumperink, and B. Nauta, "A 10-bit charge-redistribution ADC consuming 1.9 μ W at 1 MS/s," *IEEE J. Solid-State Circuits*, vol. 45, no. 5, pp. 1007–1015, May 2010.
- [13] H. S. Bindra, C. E. Lokin, D. Schinkel, A. Annema, and B. Nauta, "A 1.2-V dynamic bias latch-type comparator in 65-nm CMOS with 0.4-mV input noise," *IEEE J. Solid-State Circuits*, vol. 53, no. 7, pp. 1902–1912, Jul. 2018.

- [14] H. S. Bindra, J. Ponte, and B. Nauta, "A 174 μ V RMS input noise, 1 GS/s comparator in 22nm FDSOI with a dynamic-bias preamplifier using tail charge pump and capacitive neutralization across the latch," in *Proc. IEEE Int. Solid-State Circuits Conf. (ISSCC)*, vol. 65, Feb. 2022, pp. 1–3.
- [15] J. R. Long and M. A. Copeland, "The modeling, characterization, and design of monolithic inductors for silicon RF IC's," *IEEE J. Solid-State Circuits*, vol. 32, no. 3, pp. 357–369, Mar. 1997.
- [16] H. Wang, F. Schembari, and R. B. Staszewski, "Passive SC $\Delta\Sigma$ modulator based on pipelined charge-sharing rotation in 28-nm CMOS," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 67, no. 2, pp. 578–589, Feb. 2020.
- [17] M. Tohidian, I. Madadi, and R. B. Staszewski, "Analysis and design of a high-order discrete-time passive IIR low-pass filter," *IEEE J. Solid-State Circuits*, vol. 49, no. 11, pp. 2575–2587, Nov. 2014.
- [18] M. Tohidian, "Fully integrated discrete-time superheterodyne receiver in nano-scale CMOS," Ph.D. thesis, Dept. Electron. Instrum. Lab., Delft Univ. Technol., Delft, The Netherlands, Sep. 2015, doi: [10.4233/uuid:3c0da5c3-0ca8-490e-80d5-c942f05cc3a1](https://doi.org/10.4233/uuid:3c0da5c3-0ca8-490e-80d5-c942f05cc3a1).
- [19] B. J. Thijssen, E. A. M. Klumperink, P. Quinlan, and B. Nauta, "A 370 μ W 5.5-dB-NF BLE/BT5.0/IEEE 802.15.4-compliant receiver with >63 dB adjacent channel rejection at >2 channels offset in 22nm FDSOI," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2020, pp. 466–468.
- [20] T. Iizuka, T. Ito, and A. A. Abidi, "Comprehensive analysis of distortion in the passive FET sample-and-hold circuit," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 65, no. 4, pp. 1157–1173, Apr. 2018.
- [21] M. Dessouky and A. Kaiser, "Input switch configuration suitable for rail-to-rail operation of switched opamp circuits," *Electron. Lett.*, vol. 35, no. 1, pp. 8–10, Jan. 1999.
- [22] R. Khoini-Poorfard, L. B. Lim, and D. A. Johns, "Time-interleaved oversampling A/D converters: Theory and practice," *IEEE Trans. Circuits Syst. II, Analog Digit. Signal Process.*, vol. 44, no. 8, pp. 634–645, Aug. 1997.
- [23] D. Jiang, L. Qi, S.-W. Sin, F. Maloberti, and R. P. Martins, "A time-interleaved 2nd-order $\Delta\Sigma$ modulator achieving 5-MHz bandwidth and 86.1-dB SNDR using digital feed-forward extrapolation," *IEEE J. Solid-State Circuits*, vol. 56, no. 8, pp. 2375–2387, Aug. 2021.
- [24] A. Baluni and S. Pavan, "Analysis and design of a 20-MHz bandwidth continuous-time delta-sigma modulator with time-interleaved virtual-ground-switched FIR feedback," *IEEE J. Solid-State Circuits*, vol. 56, no. 3, pp. 729–738, Mar. 2021.
- [25] H. Wang, D. Basak, Y. Zhang, and K.-P. Pun, "A 0.59-mW 78.7-dB SNDR 2-MHz bandwidth active-RC delta-sigma modulator with relaxed and reduced amplifiers," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 68, no. 3, pp. 1114–1122, Mar. 2021.
- [26] Z. Ding, X. Zhou, and Q. Li, "A 0.5–1.1-V adaptive bypassing SAR ADC utilizing the oscillation-cycle information of a VCO-based comparator," *IEEE J. Solid-State Circuits*, vol. 54, no. 4, pp. 968–977, Apr. 2019.
- [27] S. Lee, A. P. Chandrakasan, and H. Lee, "A 12 b 5-to-50 MS/s 0.5-to-1 V voltage scalable zero-crossing based pipelined ADC," *IEEE J. Solid-State Circuits*, vol. 47, no. 7, pp. 1603–1614, Jul. 2012.
- [28] L. Somappa and M. S. Baghini, "A 300-mV auto shutdown comparator-based continuous time $\Delta\Sigma$ modulator," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 28, no. 8, pp. 1920–1924, Aug. 2020.
- [29] L. Somappa and M. S. Baghini, "A compact fully passive loop filter-based continuous time $\Delta\Sigma$ modulator for multi-channel biomedical applications," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 67, no. 2, pp. 590–599, Feb. 2020.
- [30] M. ElAnsary et al., "50nW opamp-less $\Delta\Sigma$ -modulated bioimpedance spectrum analyzer for electrochemical brain interfacing," *IEEE J. Solid-State Circuits*, vol. 55, no. 7, pp. 1971–1983, Jul. 2020.
- [31] M. ElAnsary et al., "Bidirectional peripheral nerve interface with 64 second-order opamp-less $\Delta\Sigma$ ADCs and fully integrated wireless power/data transmission," *IEEE J. Solid-State Circuits*, vol. 56, no. 11, pp. 3247–3262, Nov. 2021.
- [32] J.-E. Park, Y.-H. Hwang, and D.-K. Jeong, "A 0.4-to-1 V voltage scalable $\Delta\Sigma$ ADC with two-step hybrid integrator for IoT sensor applications in 65-nm LP CMOS," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 64, no. 12, pp. 1417–1421, Dec. 2017.



Jeroen Ponte (Graduate Student Member, IEEE) was born in Harlingen, The Netherlands. He received the B.Sc. and M.Sc. (cum laude) degrees in electrical engineering from the University of Twente, The Netherlands, in 2018 and 2020, respectively, where he is currently pursuing the Ph.D. degree with the Integrated Circuit Design Group. His research interests include but are not limited to, passive switched-capacitor filters, analog-to-digital conversion with a focus on noise-shaping architectures, and radio frequency circuits.



Harijot Singh Bindra (Member, IEEE) received the Master of Technology degree in VLSI design from Indian Institute of Technology (IIT), Delhi, in 2010, and the Ph.D. degree (cum laude) from the University of Twente, in 2019. He was a Scientist with Indian Space Research Organization (ISRO) from 2008 to 2010 and a Senior Design Engineer with Cadence Design Systems, India, from 2012 to 2014. His research interests include low-energy circuit design techniques, high-speed serial links, data converters, sensor readout circuitry, and low-power radio front-ends. He is a Technical Program Committee Member of European Solid-State Electronics Research Conference (ESSERC). He was recently awarded the VENI-2023 granted by the Dutch Research Council (NWO). He was an Associate Editor of IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—I: REGULAR PAPERS from 2021 to 2023. He was also part of the 4TU Interdisciplinary Plantenna Consortium investigating RF sensing principles for plant health monitoring.



Bram Nauta (Fellow, IEEE) was born in Hengelo, The Netherlands. He received the M.Sc. degree (cum laude) in electrical engineering and the Ph.D. degree, with a focus on analog CMOS filters for very high frequencies from the University of Twente, Enschede, The Netherlands, in 1987 and 1991, respectively.

In 1991, he joined the Mixed-Signal Circuits and Systems Department, Philips Research, Eindhoven, The Netherlands. In 1998, he returned to the University of Twente as a Full Professor heading the new IC Design Group. In 2014, he was nominated as a Distinguished Professor and from 2016 to 2020, he was the Chair of the Electrical Engineering Department. In 2022, he co-founded "ChipTechTwente," a local ecosystem initiative with 50 semiconductor-related companies, partners, and knowledge institutions. His research interests include analog and radio frequency CMOS circuits.

Dr. Nauta is a member of the Royal Netherlands Academy of Arts and Sciences (KNAW). In 2014, he received the "Simon Stevin Meester" Award (500,000 Euros), the largest national prize in The Netherlands for achievements in engineering sciences. In 2019, he received the ERC (European Research Council) Advanced Grant (2.5 Million Euros, personal grant). He was a co-recipient of the ISSCC 2002 and 2009 "Van Vessel Outstanding Paper Award" and in 2023, he received the ISSCC Author-Recognition Award for its first 70 years, as a top ten contributor. In 2023, he received the inaugural "Dutch Innovation Award" and the NWO Stevin Prize (1.5 Million Euros), the largest national science prize in The Netherlands for "exceptional success in knowledge exchange and impact for society." He was the 2013 Program Chair of the International Solid-State Circuits Conference (ISSCC). He served as the Editor-in-Chief for IEEE JOURNAL OF SOLID-STATE CIRCUITS (JSSC) from 2007 to 2010. Also, he served as an Associate Editor for IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS—II: EXPRESS BRIEFS from 1997 to 1999 and IEEE JOURNAL OF SOLID-STATE CIRCUITS from 2001 to 2006. He was on the Technical Program Committee of the Symposium on VLSI Circuits from 2009 to 2013 and served on the steering committee and program committee of European Solid-State Circuit Conference (ESSCIRC) from 1999 to 2017. He served on the program committee from 2003 to 2013 and the executive committee of ISSCC from 2007 to 2015 and since 2022. In 2023, he joined the program committee of the Advances in Analog Circuit Design Workshop Series (AACD). He was the President of the IEEE Solid-State Circuits Society from 2018 to 2019. He served two terms as a Distinguished Lecturer for the IEEE Solid-State Circuits Society.